Sessions 10–12: The relay

How to use this booklet. Each Relay Lab part has its own page, with the time, who to work with, and the steps. You may always work alone. **Lost during class?** Page 2 has the whole session on one page; every term is on page 3.

Part	What we do	Time
Opening	From one spike to the next neuron; open Relay Lab	7 min
1 Along the axon	How a spike travels; myelin; MS (Lab Part 1, page 4)	25 min
2 Across the gap	Sparks or soups? The chemical synapse; adding up (Lab Part 2, page 5)	28 min
3 Keys and locks	Transmitters, receptors, drugs (Lab Part 3, page 6)	22 min
Wrap-up	Today on one page, again; exit ticket	8 min

Warm-up

You touch a hot pan. The message travels about a metre up your arm, then passes from one neuron to the next, several times. Where would you guess the time is lost: on the long trip **along the wires**, or at the **junctions** between them? My guess:

_____

Exit ticket (end of class, 3 minutes)

1. Why does myelin make a spike faster?

_____

2. What does **Ca^{2+}** do at the axon terminal?

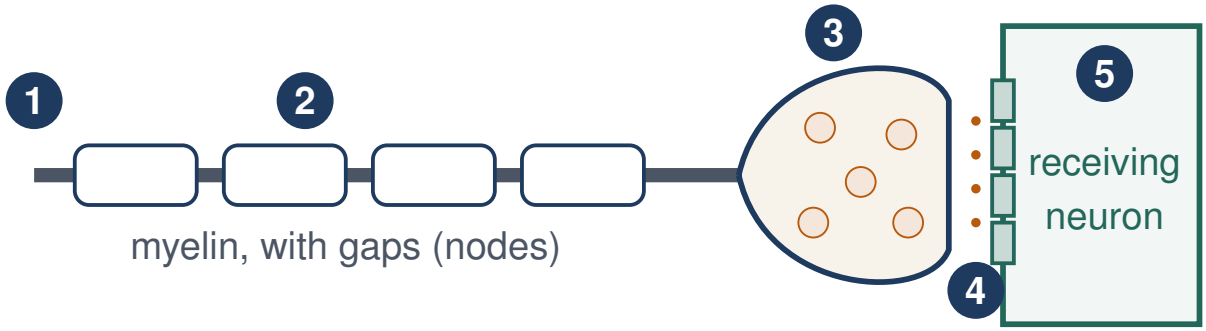
_____

3. One agonist: _____ One antagonist: _____

Next class: Quiz 2 (Sessions 7–12). Pages 2 and 8 of this booklet, and the "today on one page" sheets from Sessions 7 and 9, are good revision.

Today on one page

From one neuron to the next, in five steps. If you lose the thread, find the number here.



1. **Along the axon:** each patch fires the next, like a fuse.
2. **Myelin** makes the spike jump from gap to gap: much faster.
3. **At the terminal:** Ca^{2+} rushes in; vesicles release transmitter.
4. **Across the gap:** transmitter binds to **receptors**.
5. **The receiving neuron** adds up its inputs and decides.

Relay Lab (runs in any browser; nothing is saved or sent)

Open it on a laptop or phone and keep it beside the slides. Each part is locked until its slide (marked **RELAY LAB**) shows the password, so we move together. Part 1 is the race along the axon; Part 2 is adding up at the synapse; Part 3 is drugs at receptors.

rakesh-sengupta.github.io/neural-signalling-labs/labs/relay-lab

Word list

Keep this beside you during the lab.

conduction velocity	how fast a spike travels along an axon, in metres per second (m/s)
myelin	fatty wrapping around an axon, made by glial cells; it stops current leaking out
node of Ranvier	a short gap in the myelin, packed with **Na^+** doors, where the spike is rebuilt
saltatory conduction	the spike "jumping" from node to node (Latin *saltare*, to jump)
multiple sclerosis (MS)	a disease in which the immune system damages myelin in the brain and spinal cord
synapse	the junction where one neuron passes a signal to another
electrical synapse	a gap junction: ions flow straight through from cell to cell (gap about 3 nm)
chemical synapse	a synapse where a chemical crosses a gap of 20–50 nm
presynaptic / **postsynaptic**	the sending side / the receiving side of a synapse
vesicle	a tiny bubble (about 50 nm) filled with transmitter
exocytosis	a vesicle fusing with the membrane and releasing its contents
neurotransmitter	the chemical messenger released at a synapse
receptor	a protein that the transmitter binds to, like a key in a lock
EPSP / **IPSP**	a small push towards threshold (excitatory) / away from it (inhibitory)
summation	adding up inputs: *spatial* (several at once) or *temporal* (one, repeated fast)
ion-channel receptor	a receptor that is itself a door: fast (milliseconds)
G-protein receptor	a receptor that works through messengers inside the cell: slower, longer, amplified
agonist / **antagonist**	a drug that switches a receptor on / one that blocks it
reuptake	transmitter pumped back into the terminal, ending the signal

Relay Lab, Part 1: the race

Time: 8 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

☐ Open Relay Lab (page 2) and press **Continue**. Type the password from the slide and press **Unlock**.

☐ Choose an axon and press **Send a spike**. It races 10 cm between two electrodes.

☐ Race all four axons in the table below.

☐ Type each speed into the lab and press **Check**.

Predict first. Which will be faster: the squid giant axon (500 µm thick, bare) or a 12 µm myelinated axon from your arm? ☐ squid ☐ myelinated

Axon	Time for 10 cm	Speed	Time for 1 m
1 µm, bare (like a pain fibre)	ms	m/s	ms
12 µm, myelinated	ms	m/s	ms
squid giant axon, 500 µm, bare	ms	m/s	ms
12 µm, damaged myelin patch	ms	m/s	ms

Time for 1 m = time for 10 cm × 10. A metre is roughly from your toe to your spinal cord.

1. The squid axon is about 40 times thicker. Why does the thin myelinated axon still win?

2. In MS, patches of myelin are damaged. Using your last row, why might someone with MS see, feel or move more slowly?

Relay Lab, Part 2: adding up

Time: 8 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

☐ Type the password from the slide and press **Unlock**. Put the six steps of a chemical synapse in order (write them below too).

☐ Fire input A alone, then A + B, then A + B + C. Then add D (inhibitory).

☐ Fire A three times; find the **longest** gap that still makes a spike.

☐ Type your three numbers into the lab and press **Check**.

The six steps, in order (from "a spike arrives" to "transmitter is cleared"):

1 _____ 4 _____

2 _____ 5 _____

3 _____ 6 _____

What you fired	Highest rise above rest	Spike?
A alone		☐ yes ☐ no
A + B		☐ yes ☐ no
A + B + C		☐ yes ☐ no
A + B + C + D		☐ yes ☐ no

Rise from A alone (above -65 mV): _____ mV Inputs needed together: _____

Longest gap that still fires, A three times: _____ ms

1. Spatial or temporal summation? A + B + C: _____ A three times: _____

2. At a muscle, one spike always fires the muscle. Why is it useful that a brain neuron needs several inputs?

_____

_____

Relay Lab, Part 3: keys and locks

Time: 8 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

☐ Type the password from the slide and press **Unlock**. Pick a synapse and press **Release transmitter** with no drug.

☐ Try the **agonist**, the **antagonist** and the **clearance blocker**. Watch both panels.

☐ Do this for the muscle (ion channel) and for one G-protein synapse. Fill in the table.

Write the peak (no drug = 100), and how long the response lasts. Is it bigger or smaller, longer or shorter?

Drug	**ACh at a muscle (ion channel)**	**G-protein synapse: _____**
none	100, lasts _____ ms	100, lasts _____ ms
agonist		
antagonist		
clearance blocker		

Match each real drug to what it does. Write **Ag** (agonist), **An** (antagonist) or **B** (blocks clearance).

___ **Curare**: arrow-tip poison; paralyses muscles

___ **Caffeine**: stops adenosine making you drowsy

___ **Atropine**: eye drops that widen the pupil

___ **Nicotine**: switches on muscle ACh receptors

___ **Fluoxetine (Prozac)**: stops serotonin reuptake

___ **Nerve gas**: stops the enzyme that breaks down ACh

Think. Fluoxetine blocks reuptake within hours, yet its effect on mood takes weeks. What does that suggest about how it works?

Study cards

Soups or sparks? Eccles and Popper (1940s–1951)

Did neurons talk with electricity (“sparks”) or chemicals (“soups”)? John Eccles backed sparks. The philosopher Karl Popper told him a good theory should be stated so it could be proved wrong, and that he should try to refute his own. In 1951 Eccles’s own recordings did just that: inhibition needed a chemical. He switched sides, and shared the 1963 Nobel Prize. Electrical synapses do exist, but most synapses are chemical *(Valenstein, 2005; Brock, Coombs & Eccles, 1952)*.

Bernard Katz: transmitter comes in packets (1950s)

At the frog muscle, Katz saw tiny responses all of the same size: one vesicle each, a **quantum**. A spike at a muscle releases about 200 vesicles: a large, safe response. In the brain, one synapse may release a single vesicle, giving a push of a fraction of a millivolt *(Bear, Ch. 5)*.

Myelin in trouble: MS and Guillain–Barré

In **multiple sclerosis**, the immune system attacks myelin in the brain and spinal cord; signals slow or fail. A checkerboard pattern on a screen can show the slowing in the visual pathway. In **Guillain–Barré syndrome**, it attacks myelin in the nerves of the body, weakening the muscles *(Bear, Box 4.5)*.

Poisons that taught us the synapse

Botulinum toxin (spoiled canned food) stops ACh release. **Black widow venom** makes terminals dump all their ACh. α**-Bungarotoxin** (a snake venom) and **curare** (an arrow poison) block the ACh receptor. Each stops a different step *(Bear, Box 5.5)*.

Startle disease and strychnine

In **hyperekplexia**, a faulty glycine receptor weakens inhibition, so a sudden noise can make a person go stiff and fall. The poison **strychnine** blocks the same receptor, causing convulsions. Inhibition matters as much as excitation *(Bear, Box 5.6)*.

Snyder and Pert: the brain’s own morphine (1973)

Candace Pert and Solomon Snyder labelled opiate drugs with radioactivity and found they stuck to specific sites: **opiate receptors**. Why would the brain have receptors for a poppy drug? Because it makes its own: the **enkephalins** were found soon after *(Pert & Snyder, 1973; Bear, Ch. 6)*.

Transmitter cheat sheet

Three families (almost every transmitter is made from amino acids):

Family	Examples	Notes
Amino acids	glutamate, GABA, glycine	fast; glutamate usually excites, GABA and glycine usually inhibit
Amines	acetylcholine (ACh), dopamine, noradrenaline, serotonin, histamine	ACh moves your muscles; the others set mood, attention, sleep
Peptides	enkephalins, substance P	short chains of amino acids; slow, long-lasting

One key, many locks (from Bear, Table 6.1):

Transmitter	Receptor	Agonist	Antagonist
ACh	nicotinic (muscle)	nicotine	curare
ACh	muscarinic (heart)	muscarine	atropine
Glutamate	AMPA, NMDA	AMPA, NMDA	CNQX, AP5
GABA	$GABA_A$	muscimol	bicuculline
Adenosine	A type	adenosine	caffeine

Two kinds of lock:

	Ion-channel receptor	G-protein receptor
How it works	the receptor is itself a door; ions flow at once	the receptor wakes up G-proteins inside the cell; they act on doors or enzymes
Speed	milliseconds	hundreds of milliseconds to seconds
Amplification	none	one receptor can activate 10–20 G-proteins
Example	ACh at a muscle	ACh at the heart; most serotonin receptors

Answers (check after class)

Warm-up. The trip along a fast myelinated axon is quick (a metre in about 14 ms). Much of the time in a reaction goes on synapses and on neurons adding up their inputs.

Part 1. 1 µm bare: 100 ms, **1 m/s**, 1000 ms for 1 m. 12 µm myelinated: 1.4 ms, **72 m/s**, 14 ms. Squid: 4.5 ms, **22 m/s**, 45 ms. Damaged myelin: 9.6 ms, **10 m/s**, 96 ms. (1) Myelin stops current leaking out, so the spike only has to be rebuilt at the nodes and jumps between them; thickness alone is a costly way to speed up. (2) Where myelin is damaged, the spike crawls (or fails), so signals in vision, touch and movement arrive late or not at all.

Part 2. Steps: spike arrives; Ca^{2+} doors open and Ca^{2+} rushes in; vesicles fuse and release transmitter; transmitter crosses and binds to receptors; receptor channels open (EPSP or IPSP); transmitter is cleared. Highest rise: A alone about **7 mV** (threshold is 15 mV above rest), no spike; A + B about 13.5 mV, no spike; A + B + C: spike; A + B + C + D about 12.4 mV, no spike. Inputs needed: **3**. Longest gap: **7 ms**. (1) Spatial; temporal. (2) A brain neuron weighs many inputs and decides; a single noisy input does not set it off.

Part 3. Muscle: none 100 (about 8 ms); agonist already on before release, peak about 150; antagonist about 15; blocker about 210, lasting about 22 ms. Serotonin: none 100 (about 320 ms); agonist peak about 160; antagonist 15; blocker about 290, lasting about 590 ms. Heart (muscarinic): none 100 (about 270 ms); agonist about 165; antagonist 15; blocker about 305. Matching: curare An, nicotine Ag, caffeine An, fluoxetine B, atropine An, nerve gas B. Think: blocking reuptake cannot be the whole story. The slow effect suggests the brain itself changes over weeks in response to the extra serotonin; exactly how is still being studied.

Exit ticket. (1) The spike only has to be rebuilt at the nodes and jumps between them. (2) Ca^{2+} rushes in and makes vesicles fuse and release transmitter. (3) For example, agonist: nicotine; antagonist: curare, atropine or caffeine.

Sources

- Bear, M. F., Connors, B. W., & Paradiso, M. A. (2016). *Neuroscience: Exploring the Brain* (4th ed.), Chapters 4, 5 and 6 (Boxes 4.5, 5.5, 5.6; Table 6.1).
- Borck, C. (2017). Soups and sparks revisited. *Nuncius* 32, 286–329.
- Brock, L. G., Coombs, J. S., & Eccles, J. C. (1952). The recording of potentials from motoneurones with an intracellular electrode. *Journal of Physiology* 117, 431–460.
- de Leeuw, J. R. (2015). jsPsych: a JavaScript library for creating behavioral experiments in a web browser. *Behavior Research Methods* 47, 1–12.
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- Purves, D., et al. (Eds.) *Neuroscience*. Sinauer. Fibre classes and conduction velocities.
- Valenstein, E. S. (2005). *The War of the Soups and the Sparks*. Columbia University Press.
- Relay Lab, Part 1: conduction speeds from rules of thumb fitted to real nerves (myelinated: about 6 m/s per µm of diameter, after Hursh 1939; bare: about $\sqrt{d}$ m/s), not a simulation. The damaged patch covers 3 of the 10 cm.
- Relay Lab, Part 2: a leaky integrate-and-fire neuron with conductance-based synapses (membrane time constant 15 ms; rest -65 mV; threshold -50 mV).
- Relay Lab, Part 3: simplified receptor kinetics in arbitrary units, scaled so that no drug $=$ 100. The directions of the effects are what matter, not the exact numbers.